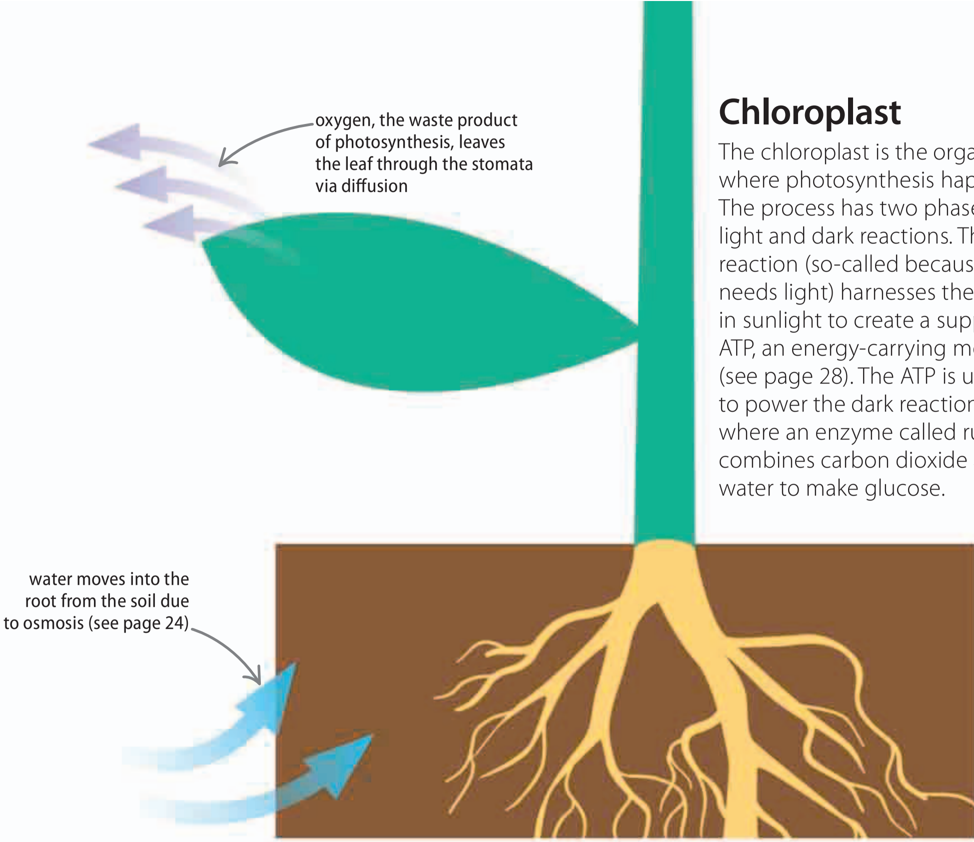
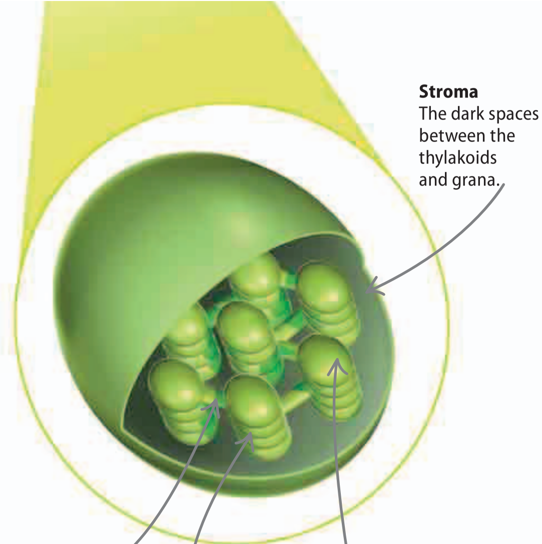


Chloroplast

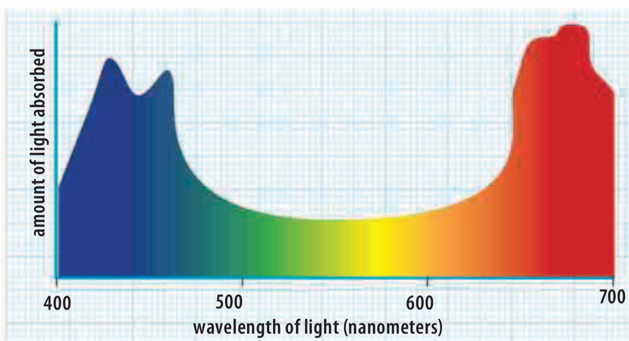
The chloroplast is the organelle where photosynthesis happens. The process has two phases, the light and dark reactions. The light reaction (so-called because it needs light) harnesses the energy in sunlight to create a supply of ATP, an energy-carrying molecule (see page 28). The ATP is used to power the dark reaction, where an enzyme called rubisco combines carbon dioxide and water to make glucose.



△ **Inside a chloroplast**
The chlorophyll molecules are attached to membranes called thylakoids. The dark reaction takes place in the stroma, the spaces between the thylakoids and grana. All green parts of a plant contain cells filled with chloroplasts.

Chlorophyll

The chemical pigment chlorophyll is what makes most plants look green. Each chlorophyll molecule absorbs the red and blue light in sunlight, using its energy to power photosynthesis, and reflects the rest back. So what we see is the green light that is not used by photosynthesis reflected back.



△ **Absorption spectrum**
This graph shows the wavelengths, or colors, of light, that are absorbed by chlorophyll. The dip in the middle shows that yellows and greens are absorbed less than reds and blues.

REAL WORLD

Fall colors

Deciduous trees drop their leaves in winter, when it is too dark to photosynthesize efficiently. Before they are shed, the leaves change color—turning from green to brown. This change is due to the chlorophyll being absorbed by the plant for use in the next year. The fall colors are formed by pigments called carotenes that are left behind.